



Momentum and Impulse · Mark Scheme

Further Mechanics 1 Chapter 1: Momentum and Impulse. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Momentum and impulse

- 1 $I = 8 \times 3 = 24 \text{ N s}$ B1; $2.5v = 15 + 24$, $v = 15.6 \text{ ms}^{-1}$ B1.
- 2 Taking the rebound direction as positive: $I = 0.4(10) - 0.4(-15)$ M1 A1; $= 10 \text{ N s}$ A1.
- 3 $-16 = 2v - 2(5)$ M1 A1; $v = -3$, so speed 3 ms^{-1} in the opposite direction to its original motion A1.
- 4 $v^2 = 2(9.8)(3.6)$ and $w^2 = 2(9.8)(2.5)$ M1; $v = 8.4 \text{ ms}^{-1}$, $w = 7 \text{ ms}^{-1}$ A1; $I = 0.25(7) - 0.25(-8.4)$ M1; $= 3.85 \text{ N s}$ vertically upwards A1.
- 5(a) Positive in the new direction: $J = 0.5(6) - 0.5(-4)$ M1 A1; $J = 5$ A1.
- 5(b) $F \times 2.5 = 0.5(6) - 0$ M1; $F = 1.2 \text{ N}$ A1; $R = 0.5(9.8) = 4.9$, $\mu = \frac{1.2}{4.9}$ M1; $\mu = 0.245$ A1.
- 5(c) e.g. P is a particle / friction is the only horizontal force (no air resistance) / friction is constant B1.

Section B – Conservation of momentum

- 6 $4(6) + 0 = 6v$ M1; $v = 4 \text{ ms}^{-1}$ A1.
- 7 Positive in A 's direction: $3(5) + 2(-2) = 3(1) + 2v$ M1 A1; $v = 4$, so B moves at 4 ms^{-1} in A 's original direction A1.
- 8 $0 = 0.02(300) - 1.2V$ M1; $V = 5 \text{ ms}^{-1}$ A1; impulse on bullet $= 0.02(300) = 6 \text{ N s}$ B1.
- 9 $2(10) = (2 + 3)v$ M1; $v = 4 \text{ ms}^{-1}$ A1; impulse on B : $I = 3(4) - 0$ M1; $= 12 \text{ N s}$ A1.
- 10(a) Positive in P 's direction: $2(5) + 1(-4) = 2v + 1(3v)$ M1 A1; $v = 1.2$ A1; P : 1.2 ms^{-1} , Q : 3.6 ms^{-1} A1.
- 10(b) $I = 1(3.6) - 1(-4)$ M1; $= 7.6 \text{ N s}$ A1.
- 10(c) $1(3.6) + 2(0.6) = 3w$ M1; $w = 1.6 \text{ ms}^{-1}$ A1; P moves at $1.2 \text{ ms}^{-1} < 1.6 \text{ ms}^{-1}$ in the same direction, so P does not collide with it again B1.

Section C – Two dimensions

- 11 $(6\mathbf{i} + 8\mathbf{j}) \text{ N s}$ B1; $\sqrt{6^2 + 8^2} = 10 \text{ N s}$ B1.
- 12 $(3\mathbf{i} + 5\mathbf{j}) = 0.5\mathbf{v} - 0.5(4\mathbf{i} - 2\mathbf{j})$ M1; $\mathbf{v} = (10\mathbf{i} + 8\mathbf{j}) \text{ ms}^{-1}$ A1; speed $\sqrt{164} = 12.8 \text{ ms}^{-1}$ A1.
- 13 $\mathbf{I} = 3(-4\mathbf{i} + \mathbf{j}) - 3(2\mathbf{i} + 5\mathbf{j})$ M1; $= (-18\mathbf{i} - 12\mathbf{j}) \text{ N s}$ A1; $|\mathbf{I}| = \sqrt{468} = 21.6 \text{ N s}$ A1.
- 14 $\mathbf{I} = 0.2(20\mathbf{j}) - 0.2(15\mathbf{i}) = -3\mathbf{i} + 4\mathbf{j}$ M1 A1; $|\mathbf{I}| = 5 \text{ N s}$ A1; angle $= 180^\circ - \tan^{-1} \frac{4}{3} = 127^\circ$ A1.
- 15(a) $14 \cos 60^\circ = 7$, $14 \sin 60^\circ = 7\sqrt{3}$ M1; $(7\mathbf{i} + 7\sqrt{3}\mathbf{j}) \text{ ms}^{-1}$ A1.
- 15(b) $\mathbf{J} = 0.4(7\mathbf{i} + 7\sqrt{3}\mathbf{j}) - 0.4(10\mathbf{i})$ M1; $= (-1.2\mathbf{i} + 2.8\sqrt{3}\mathbf{j}) \text{ N s}$ A1; $|\mathbf{J}| = \sqrt{1.44 + 23.52} = \sqrt{24.96} = 4.996 = 5.00 \text{ N s}$ A1.
- 15(c) $\tan^{-1} \left(\frac{2.8\sqrt{3}}{1.2} \right) = 76.1^\circ$ M1 A1; $\mathbf{J}$ has a negative $\mathbf{i}$ -component, so the angle is $180^\circ - 76.1^\circ = 104^\circ$ A1.

Roar-level challenge

- 16 $v_B^2 = 2(9.8)(0.9)$, $v_B = 4.2 \text{ ms}^{-1}$ B1; for B : $3(4.2) - J = 3v$ M1; for A : $J = 2v$ M1; $12.6 = 5v$, $v = 2.52 \text{ ms}^{-1}$ A1; $J = 5.04 \text{ N s}$ A1.
(Weights are ignored during the jerk: the impulse is instantaneous.)